

Chapter 2, Problem 3P

Problem

A bar of silicon is 4 cm long with a circular cross section. If the resistance of the bar is $240\ \Omega$ at room temperature, what is the cross-sectional radius of the bar?

Step-by-step solution

Step 1 of 4

Consider the following data:

Length of the **silicon** bar with **circular cross** section is 4 cm or 0.04 meters

At room temperature resistance of the silicon bar is $240\ \Omega$

Step 2 of 4

Represent the resistance in mathematical form.

$$R = \rho \frac{l}{A} \dots\dots (1)$$

Where R is resistance and

l is the length

A is the cross sectional area of physical property (in this problem its silicon bar).

Step 3 of 4

From the Resistivities of common materials TABLE 2.1 it is clear that the resistivity of the silicon material is 6.4×10^2 ohm-meters or $\Omega \cdot \text{m}$.

Substitute 6.4×10^2 for ρ , $240\ \Omega$ for R and 0.04 m for l in equation (1).

$$240 = (6.4 \times 10^2) \frac{(0.04)}{A}$$

Rewrite the equation.

$$\begin{aligned} A &= (6.4 \times 10^2) \frac{(0.04)}{240} \\ &= 0.1067\ \text{m}^2 \end{aligned}$$

Step 4 of 4

Write the formula for the circular cross sectional area of silicon bar.

$$A = \pi r^2$$

Substitute 0.1067 for A in the above equation.

$$\pi r^2 = 0.1067$$

$$\begin{aligned} r^2 &= \frac{0.1067}{\pi} \\ &= \frac{0.1067}{3.14} \quad \text{since } \pi = 3.14 \\ &= 0.03395 \end{aligned}$$

Further simplified

$$\begin{aligned} r^2 &= 0.03395 \\ &= \sqrt{0.03395} \\ &\cong 0.1843 \text{ m} \\ &\cong 184.3 \text{ mm} \end{aligned}$$

Therefore the cross sectional radius of the bar is 184.3 mm.